M2 Findings Memo – Ride or Rail?

CS 483 Big Data Mining · Team Loop Miners (Harsh Shah, Dharmik Patel, Yousuf Hussain) · Milestone 2: Patterns found

Summary

Most Chicago rideshare trips do not fill a gap in transit: they copy routes the 'L' already serves, mostly downtown and on the North Side, in the evening and on weekends. The real gaps are smaller and more specific: busy routes that need a train transfer, and daytime trips in the South and West Side areas far from the 'L'. Three results support this, all from the full data (2023 onward, after the COVID break):

1. **K-means splits the 77 community areas into three groups** (silhouette 0.643): a dense central and lakefront group with 64% of trips, mostly evenings and weekends; the other 58 areas, with daytime everyday trips; and the two airports.
2. **30 of the 50 busiest routes between areas run along one 'L' line**, carrying 64% of those routes' trips. The busiest route without one, Near North Side ↔ Near West Side, has 10.7 million trips.
3. **35 of the 77 areas have no 'L' station**, mostly on the South and West Sides, and almost all of them fall in the daytime, everyday cluster.

Data and methods

We used all 628,189,680 trips in the City of Chicago rideshare dataset (Nov 2018 – Aug 2026), stored as Parquet in Google Cloud Storage and processed with Spark on Dataproc Serverless. Cleaning removed 582,711 trips (0.09%) with zero or extreme time or distance. Ridership dropped about 80% in April 2020 and has been flat at 7.5–8 million trips a month since 2024, so the mining below uses trips from 2023 onward only. We work at community-area level, not census tract, because 39% of trips have no tract.

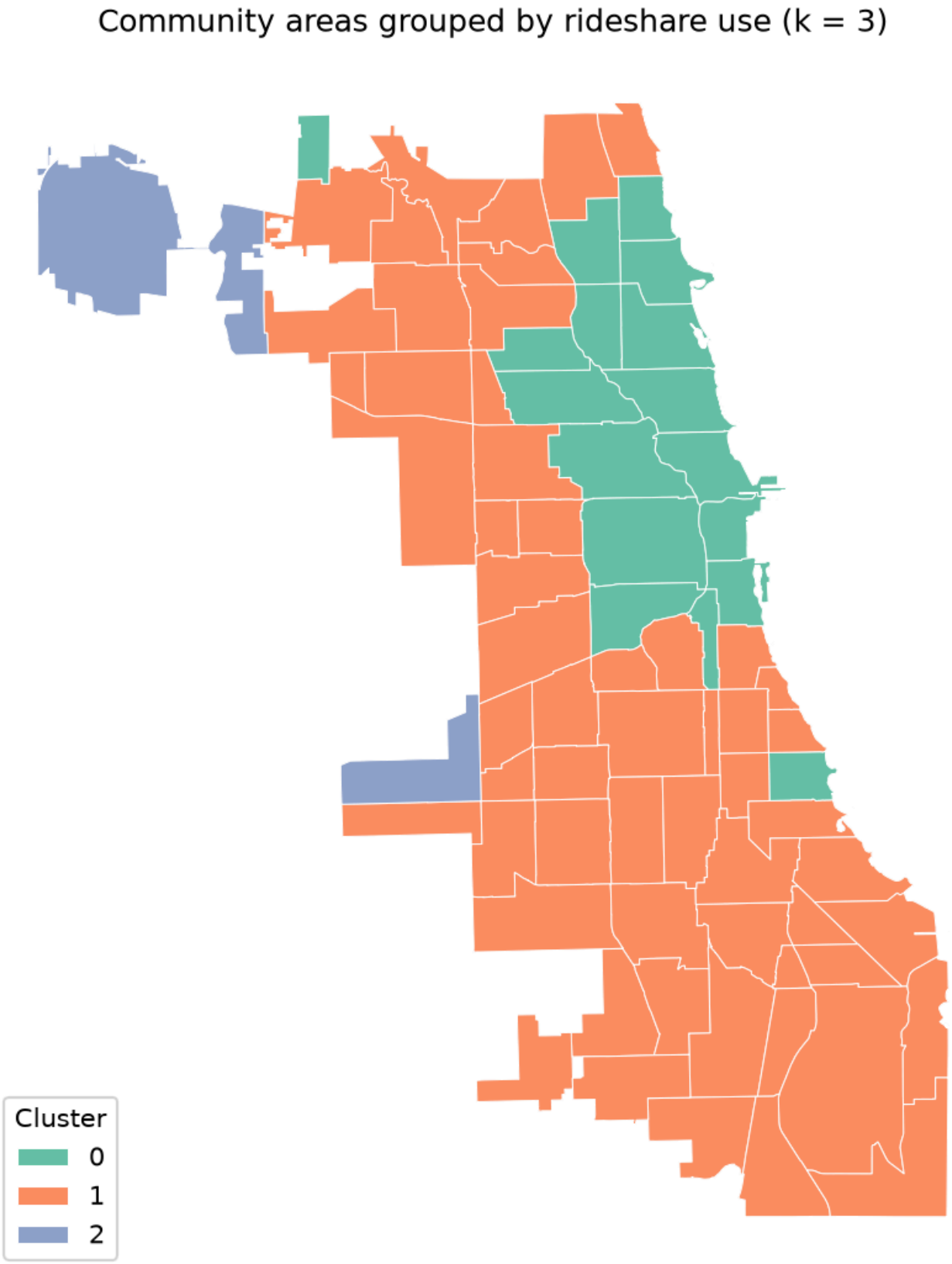
Method	Input	Settings	Evaluation
K-means clustering	One profile per community area (77 areas, 297.8M trips): share of trips in 5 time bands, weekend share, share to downtown, to the airports and within the area, average distance, average fare, trips per km²	Features standardized; k = 2 to 10	Silhouette score and elbow (within-cluster cost)
FP-Growth	One basket per trip: origin area, destination area, time band, weekday/weekend, distance band	10% random sample (26.6M baskets); min support 0.2%, min confidence 10%	Support, confidence, lift
Origin-destination counts	All 266.3M trips with both areas known	Exact counts, top 500 routes	Share of all trips
CTA access	'L' stop locations and community area boundaries	Stations and lines inside each area; distance from area center to nearest station	—

All times are Chicago local time. An early run read the timestamps as UTC, which shifted every time band by 5–6 hours; we found it because 23–35% of trips appeared to start between midnight and 6 AM, fixed it, and reran both jobs.

Finding 1: three kinds of rideshare areas

K-means with k = 3 gives the clearest grouping (silhouette 0.643; k = 2 scores 0.579 and every k from 5 to 10 scores below 0.44). The three groups use rideshare in very different ways:

Cluster	Areas	Trips (2023+)	What makes it different
0 – Central and lakefront	17: Loop, Near North, Near West, Lake View, Lincoln Park, West Town, Logan Square, Hyde Park and others	189.9M (64%)	Most evening (25%), late-night (23%) and weekend (36%) trips; 32% go downtown; highest trips per km²
1 – Rest of the city	58: most of the South, West and Northwest Sides	87.1M (29%)	Most morning (20%) and midday (33%) trips; only 14% go downtown; lower density
2 – Airports	2: O'Hare and Garfield Ridge (Midway)	20.8M (7%)	Much longer and more expensive trips (4.9 and 5.6 standard deviations above average); 47% go downtown

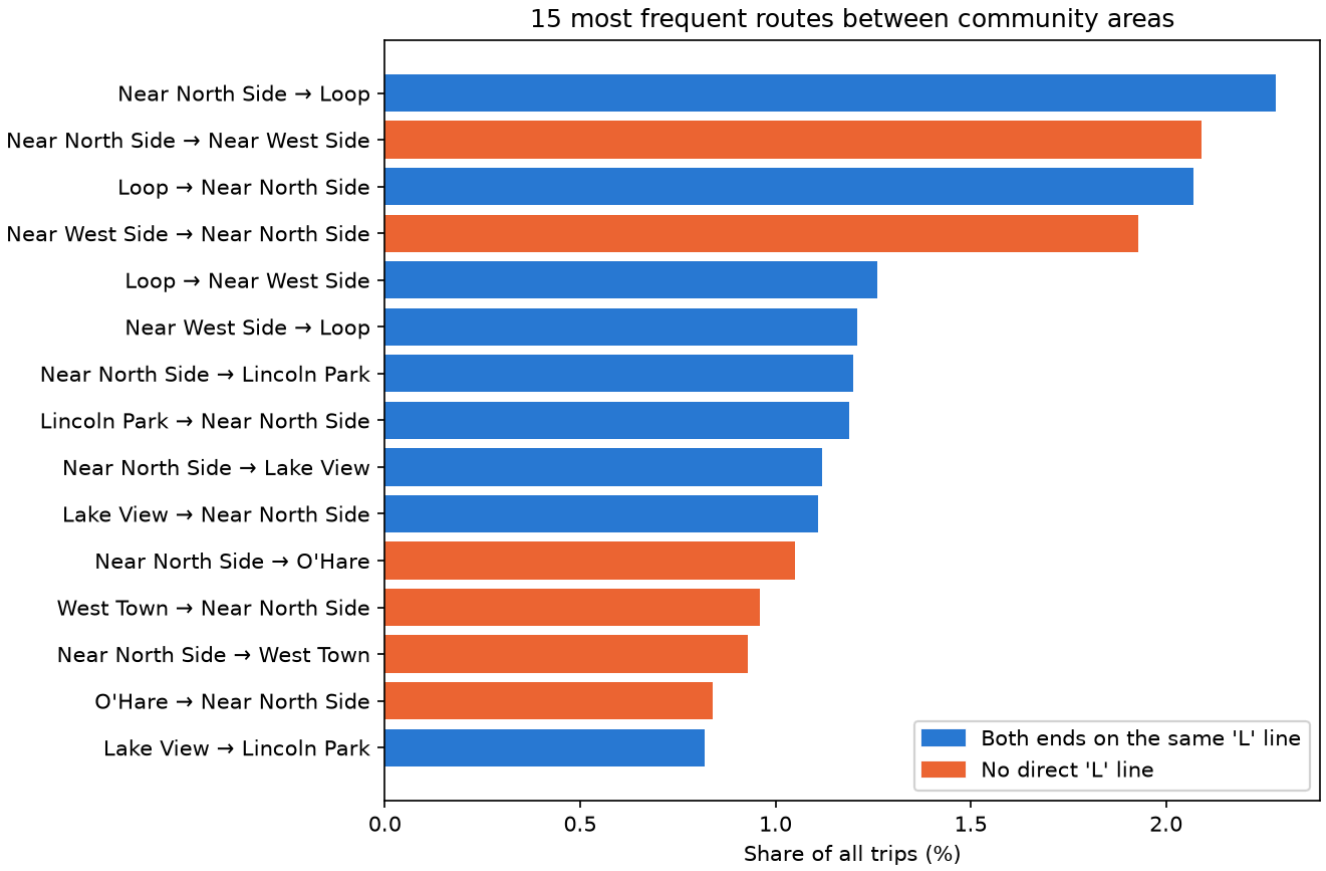

Community areas grouped by rideshare use (k = 3)
Cluster
0
1
2

Cluster 0 follows the North Side 'L' lines, so its heavy rideshare use is mostly about convenience at night and on weekends. Cluster 1 is where everyday daytime trips happen, and it is where we will look for transit gaps.

Limitations: K-means starts from random centers, so the cost is not always lower for a larger k (it rises from k = 4 to k = 5). Edison Park lands in Cluster 0 with only 324,000 trips, so its profile is less reliable.

Finding 2: most busy routes copy an 'L' line

Rideshare is very concentrated: the 20 busiest routes carry 28.5% of all trips, and the single busiest is a trip that stays inside the Near North Side (10.2M trips, 3.8%). Of the 50 busiest routes between different areas, 30 can be done on one 'L' line with no transfer, and they carry 64% of those routes' trips. Loop ↔ O'Hare alone has 3.4M trips even though the Blue Line runs directly between them.



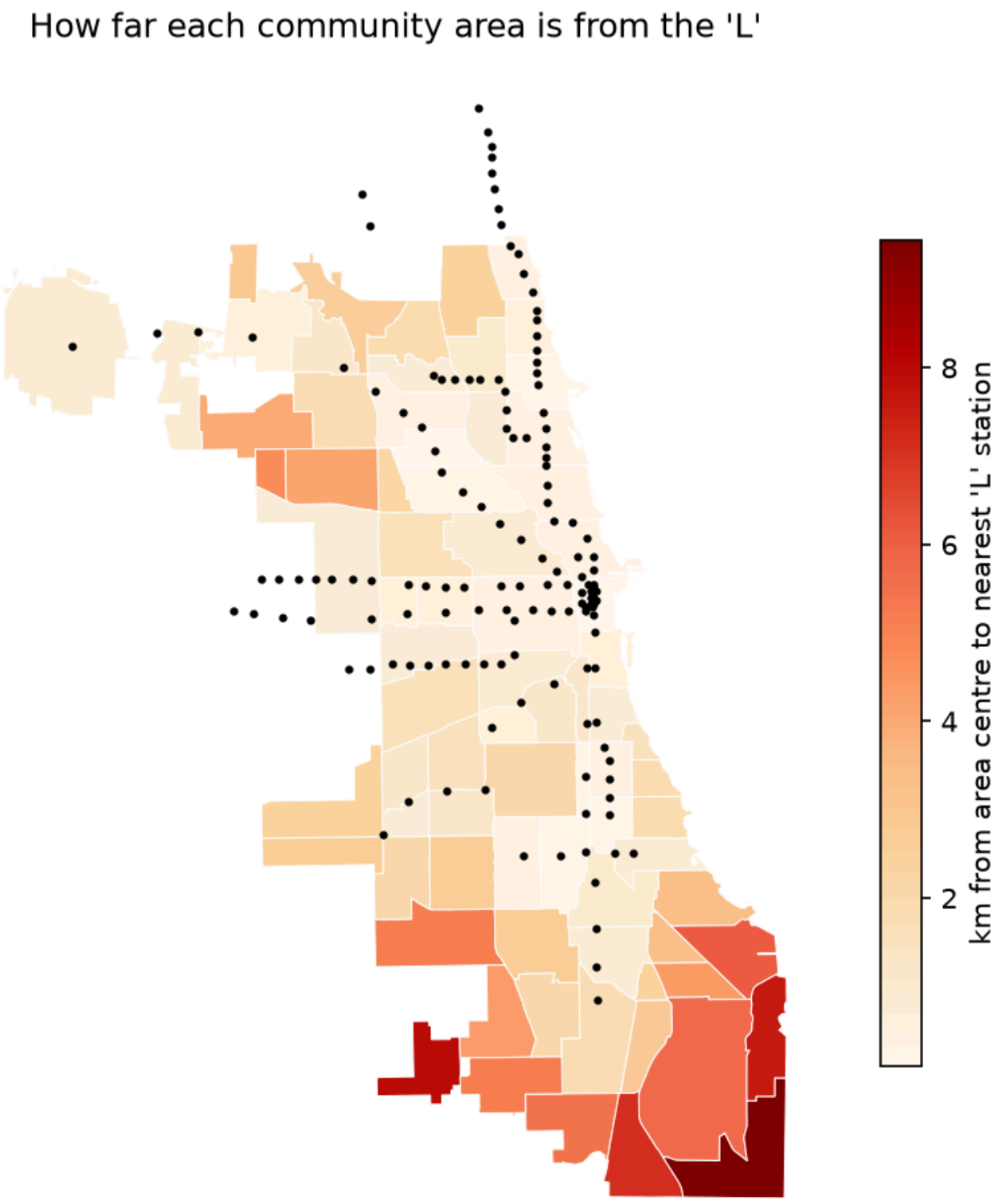
The busiest routes that need a transfer are our first transit-gap candidates:

Route (both directions)	Trips (2023+)	Share of all trips	Why no direct 'L'
Near North Side ↔ Near West Side	10.7M	4.0%	Red/Brown/Purple vs Blue/Green/Pink: transfer needed
Near North Side ↔ O'Hare	5.1M	1.9%	O'Hare is Blue only
Near North Side ↔ West Town	5.0M	1.9%	West Town is Blue only

FP-Growth adds when and how these trips happen. If a trip starts in the Near North Side on a weekday morning and is longer than 10 miles, it goes to O'Hare 70% of the time (confidence 0.698), 15.8 times more often than an average trip (lift). Short weekend trips from Lake View are 2.5 times more likely than average to be at night (12–6 AM), and 71% of night trips to Lincoln Park are on weekends, matching the nightlife pattern of Cluster 0. Rules inside one area have the highest lift (Austin, 56; Hyde Park, 32, mostly evening trips), partly because each single area is a rare item, so we read lift together with support and confidence.

Finding 3: where the 'L' doesn't reach

35 of the 77 community areas have no 'L' station inside them, and 33 of those 35 are in Cluster 1. The median distance from an area's center to the nearest station is 0.5 km in Cluster 0 but 1.9 km in Cluster 1. The farthest areas are all on the Far South Side: Hegewisch (9.4 km), Mount Greenwood (8.0 km), East Side (7.6 km), Riverdale (7.1 km) and South Chicago (6.1 km).



So the areas with the most rideshare trips have the best 'L' access, while the areas with the worst access have fewer, mostly daytime trips. For those areas, rideshare may be filling a real gap, and trips per person will tell us more than total trips.

Progress, difficulties and next steps

Milestones 1 and 2 are done, ahead of the sprint dates.

Milestone	Planned	Status
M1 Data ready: all trips as Parquet, data-quality report, EDA	Oct 30	Done Nov 1 (full data: 628,189,680 trips)
M2 Patterns found: feature table, CTA access, K-means, FP-Growth, memo	Nov 13	Done Nov 6
M3 Model + prototype: demand model, transit-gap score, dashboard	Week 13–14	Next sprint
M4 Final report and presentation with demo	Week 15 (presentation Dec 1/3, report Dec 6)	Planned

Difficulties and how we solved them:

Problem	What we did
156 GB of CSV did not fit on one laptop during conversion	Converted one table at a time and deleted each CSV after the row counts matched exactly
The 2018–2022 table has 3 fewer columns than the newer tables	Read all tables with mergeSchema, so missing columns are empty
Full-data Spark jobs crashed laptops	Moved heavy jobs to Google Cloud Dataproc Serverless (free trial); laptops only for small tests
The free trial allows one Dataproc job at a time (12 CPUs)	Run jobs one after another and check that the previous one has released its resources
Timestamps read as UTC on Dataproc, shifting time bands by 5–6 hours	Set the Spark session to Chicago time and reran both jobs
Our first 'L' check only used each area's nearest station and missed lines	Recomputed using every station inside each area

Next steps (Sprint 3, Nov 16–24):

- Demand model: gradient-boosted trees predicting trips per area and hour from Dharmik's feature table (5.26M rows, with weather and CTA ridership), compared with a simple baseline using RMSE and MAE on a time-based split.
- Transit-gap score per area: trips per person (census population), distance to the 'L', transfers needed on its busiest routes, and income.
- Dashboard prototype (Plotly Dash): a map of the transit-gap score and demand by hour, for the Week 15 demo.

Who did what (Sprints 1–2):

Member	Work
Harsh Shah	Repository, board and Google Cloud setup; Parquet conversion script; sample data-quality notebook; K-means and FP-Growth on Dataproc and their analysis notebooks; this memo

Member	Work
Dharmik Patel	Downloaded and verified all 628M trips; ran the Parquet conversion; full-data quality and trip EDA notebooks; uploaded the data to Cloud Storage; area × hour feature table
Yousuf Hussain	CTA, weather, census and boundary data; supporting EDA; CTA access per area with maps; Scrum Master for Week 11 (Sprint 2 kickoff, stand-ups, milestone dates on Piazza)

Code, notebooks and figures: github.com/harshsha0703/ride-or-rail (`src/mining/` , `notebooks/04` – `06` , `docs/figures/`).